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Job creation in tight and slack labor markets☆

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ABSTRACT

Does investment create more jobs in slack than in tight labor markets? We study this question using data from a photovoltaic investment scheme. Comparing counties with high and low unemployment over time and across space, we find that € 100,000 of investment created 1.2 job-years in slack markets and fewer than 0.5 job-years in tight markets. This corresponds to labor earnings multipliers of 1.1 and below 0.5, respectively. These differences are not driven by changes in investment composition, capital-labor substitution, or regional migration. Consistent with crowding-out as a mechanism, investment leads to higher wage growth in tight than in slack markets.

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1. Introduction

The Great Recession has sparked renewed interest in the effects of stimulus spending. One focus of the debate is whether the size of the multiplier depends on the state of the economy, i.e., whether additional demand shocks have larger effects on output in times of slack resources than in times of full capacity utilization. This question is still unresolved: The growing empirical literature using either variation on the national level over time or cross-sectional variation on the regional level has not come to a consensus yet.[1]

Assessing the state-dependence of the multiplier is challenging for three reasons: First, identifying the multiplier requires exogenous demand shocks. Second, these shocks need to vary sufficiently within each state of the economy. Otherwise there is potentially not enough statistical power to estimate the multiplier for each state. Third, the shocks have to be comparable in their composition across the different states.

The literature to date has addressed the first and second challenge. Shocks that are plausibly exogenous to the economic circumstances have been identified, for example, from structural VARs, Ramey's (2011) military news, and a variety of in-

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[1] For example, Auerbach and Gorodnichenko (2012, 2013), Bachmann and Sims (2012), Fazzari et al. (2014), Nakamura and Steinsson (2014), Shoag (2015), and Dube et al. (2018) find evidence for state dependent multipliers, while others find little differences in the multiplier by the state of the economy (e.g., Biolsi, 2017; Caggiano et al., 2015; Owyang et al., 2013; Ramey and Zubairy, 2018).

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strumental variables on the subnational level.[2] In order to have a sufficient amount of shocks across different states of the economy, prior work has extended time series data to the late 19th century (Ramey and Zubairy, 2018), combined data from multiple countries (Auerbach and Gorodnichenko, 2013), and focused on the regional level (Dube et al., 2018). In contrast to addressing the issues of exogeneity and statistical power, the literature has been silent about the third challenge: a differential composition of spending across the states of the economy. Specifically, one issue could be that the primary goal of public investments in booms is to increase the long-run growth potential with projects characterized by long planning horizons. Spending during recessions, in contrast, may be designed to deliver a quick boost to the economy. In this case, the state-dependent composition of demand may lead to differential short-run effects on economic activity, even for demand shocks of the same size.

In this paper, we are the first to tackle all three challenges at once. We do so by studying local demand shocks from investment in rooftop photovoltaic (PV) systems in Germany between 2003 and 2012. In this time period, over € 60 billion were invested in household PV systems due to a guaranteed price at which the owners of such systems could sell their produced electricity. Using the cross-sectional and time-series variation resulting from this program, this paper seeks to answer the following question: Did PV installations create more jobs in slack than in tight labor markets?

The setting of PV investment in Germany is ideal to estimate state-dependent multipliers as PV installations constituted an exogenous, frequent, and comparable demand shock in all states of the economy. First, the variation in PV installations over time and across space was mainly driven by factors that are exogenous to the economic circumstances: Over time, investment was largely determined by the world market price of solar modules in relation to the amount of the guaranteed price. The spatial variation in investments across the 400 German counties depends on the local solar radiation and the availability of suitable rooftops. Second, there is ample identifying variation in these investments for any partition of counties into groups with slack and tight labor markets, as the installation of rooftop PV systems was profitable in all German regions. Third, the composition of investment has been constant as each PV installation of a given size constituted the same demand shock. Also, we directly observe the physical amount of investments. Hence, variation in regional prices, which may be a function of labor market tightness, cannot affect our results.

Our main finding is that the installation of PV systems created at least twice as many jobs in slack labor markets characterized by high unemployment than in tight labor markets with low unemployment. Our preferred specifications compare job creation in tight and slack markets using two sample splits: First, along the time series dimension, we compare counties at times when their unemployment rate is above or below its long-run average. Second, along the cross-sectional dimension, we compare counties with high or low unemployment relative to their state average in a given year. For these sample splits, the installation of PV systems with a capacity of one megawatt peak (MWp) led to about 37 new jobs in slack labor markets, but created only around 13 jobs or less in tight labor markets. This implies that € 100,000 of investment created 1.2 job-years in slack and fewer than 0.5 jobs in tight markets. Moreover, using these estimates to approximate a local labor earnings multiplier, we arrive at a multiplier of 1.1 in slack markets and below 0.5 in tight markets.[3] These results are robust to various alternative ways of classifying slack and tight markets, and remain qualitatively unchanged when we instrument PV installations with their profitability as measured by the investments' net present value.

Furthermore, we make progress on identifying the mechanism underlying the state dependence of the effects. For example, higher crowding out in tight labor markets may explain our findings, consistent with the model of Michaillat (2014).[4] Additional demand for PV installations might draw workers from other jobs and put upward pressure on wages in tight markets, while it creates new jobs in slack markets. Another mechanism could be that in a tight market, companies substitute labor for capital. For example, they might use a machine instead of labor to move solar panels to roofs. Lastly, in slack markets additional demand in a county might be accommodated by an increase in employment in the same county, while in tight markets it might lead to an increase in employment mainly in other regions.

The evidence is most consistent with the crowding-out mechanism put forward by Michaillat (2014). First, a county's employment is largely unaffected by additional demand in surrounding regions, irrespective of the state of the county's labor market. This excludes regional spillovers as a mechanism. Second, the differential employment gains in slack and tight markets are driven both by the directly exposed sector installing PV systems as well as indirectly exposed local service sectors. This rules out a simple substitution of labor and capital. Finally, consistent with crowding-out, there is evidence that investments lead to additional wage growth in the construction sector in tight markets, but not in slack markets.

[2] Examples for instrumental variables used in the literature are windfall profits from state pension plans (Shoag, 2015), exposure to military spending (Nakamura and Steinsson, 2014), or shocks to population counts from the decennial census (Suárez Serrato and Wingender, 2016).

[3] Abusing terminology slightly, a local labor earnings multiplier of x means that a *local* investment of the yearly gross median wage delivers x additional local jobs. This figure accounts for the fact that of the total costs of installing rooftop PV systems only one third accrue locally, while two thirds are spent on tradable parts (solar panels and components).

[4] In the search and matching framework of Michaillat (2014), higher crowding out in tight markets is a consequence of a convex quasi-labor supply function. Roulleau-Pasdeloup (2017) suggests a similar mechanism based on search frictions and downward-rigidity in prices. Recent work by Rendahl (2016) also emphasizes the role of the labor market for generating differential effects of fiscal spending in booms and recessions, albeit with a different channel. The mechanism in this paper relies on differential expectations regarding the path of the entire economy at and away from the zero lower bound. For most of our sample these differential expectations are likely not present and, hence, cannot explain the results.

Taken together, we provide evidence that the employment effects of PV investment in Germany are state-dependent and that crowding-out is the mechanism behind this finding. Yet, it is important to keep in mind that we estimate local multipliers from relative employment gains across regions. In consequence, our estimates do not account for the general equilibrium effects of monetary policy, Ricardian equivalence, and regional spillovers. For this reason, local multipliers tend to be larger than the national multiplier in theory (Farhi and Werning, 2016; Nakamura and Steinsson, 2014), and converting local into national multipliers requires additional assumptions (see, e.g., Chodorow-Reich, forthcoming).[5]

This paper is thus complementary to the time-series literature on the state-dependence of multipliers (Auerbach and Gorodnichenko, 2012; 2013; Bachmann and Sims, 2012; Biolsi, 2017; Caggiano et al., 2015; Fazzari et al., 2014; Ramey and Zubairy, 2018). Specifically, in addition to this paper, only Auerbach and Gorodnichenko (2013) investigate potential channels for state-dependence. In a cross-country setting, they show that at times of low output growth, employment responds stronger to demand shocks while wage growth is muted. The opposite is true at times of high output growth, also pointing towards crowding-out as a likely mechanism.

This paper adds more directly to the literature on local multipliers. So far, this literature is primarily concerned with estimating the unconditional multiplier.[6] Our paper and recent work by Dube et al. (2018) are the first to focus on its state-dependence, although some papers evaluate the state-dependence of their estimates in auxiliary analyses (Adelino et al., 2017; Brückner and Tuladhar, 2014; Cohen et al., 2011; Nakamura and Steinsson, 2014; Shoag, 2015; Suárez Serrato and Wingender, 2016).[7] The majority of these papers find—as we do—that multipliers are larger if production inputs are slack.

In relation to these papers, we make progress on three fronts. First, this paper is unique in ruling out price and composition effects as potential confounding factors for state-dependent multipliers. Ruling out local price effects is important, as otherwise the same nominal amount of spending may constitute a different real shock. Ruling out composition effects is important, as the state of local economies may directly affect the composition of public funds.[8] Second, we are the first to provide evidence for crowding-out as the mechanism for larger multipliers in slack than in tight labor markets. Third, we demonstrate the robustness of our results with respect to a wide array of different ways of classifying slack and tight markets.

While PV installations and their constant composition are ideal to test the state-dependence of the multiplier, it is unclear whether they are an effective tool to stimulate local demand: The scope for installing PV systems is limited; once one system is mounted to the roof there is little economic rationale to add another one. Furthermore, two thirds of the investment costs accrue to the tradable system components which are mostly produced overseas. Nevertheless, to the extent that our findings are informative about the job gains from small-scale construction activity, they suggest that such undertakings may be a viable stimulus at times of slack (see also Buchheim and Watzinger, 2017).

The remainder of this paper is structured as follows. Section 2 describes the institutional background and the data. We lay out the empirical approach in Section 3, where we also discuss the identifying variation in PV installations. Section 4 reports the main results. Section 5 discusses potential mechanisms and Section 6 concludes.

2. Institutional background and data

The German Renewable Energy Act (*Gesetz für den Vorrang Erneuerbarer Energien*) went into effect on April 1st 2000 with the aim to increase the share of renewable energy in German energy production. The current target is that 80% of German electricity consumption stems from renewable energy sources by 2050. In order to achieve this, the law rests on two key mechanisms: First, the law mandates grid operators to connect all (household) renewable energy systems to the grid and to purchase the produced electricity. Second, this electricity is remunerated with a guaranteed feed-in tariff specified in the law. The relevant feed-in tariff for a given source of renewable energy is determined at the time at which it is connected to the grid and remains fixed for 20 years thereafter. In other words, for an existing renewable power plant (e.g., a PV system), the feed-in tariff cannot be changed retroactively. The feed-in tariff is independent of the market price for electricity, but has always been considerably higher than the market price during the sample period.

2.1. Physical investments in rooftop PV systems

An additional provision of the Renewable Energy Act is that until July 2014 it mandated grid operators to collect and publish data on all renewable energy power plants. The data provided by the grid operators has been aggregated, cleaned, and validated by the *Deutsche Gesellschaft für Sonnenenergie* (DGS), which is the German branch of the International Solar

[5] See, however, Dupor and Guerrero (2017), who show empirically that the difference between the local and aggregate multipliers of military spending is small.

[6] See Fuchs-Schündeln and Hassan (2016) and Chodorow-Reich (forthcoming) for recent reviews.

[7] Specifically, Dube et al. (2018) show that the projects financed by the American Recovery and Reinvestment Act during the Great Recession had a stronger effect on employment and earnings in counties hit harder by the recession.

[8] This is an important concern if local governments have discretion in how to best use available funds. This applies to the work by Shoag (2015), Suárez Serrato and Wingender (2016), and Adelino et al. (2017), who all instrument spending with changes in the local government's budget constraints.

Energy Society. We use the data up to and including 2012 as data postings have become more sketchy in 2013 ahead of the change in the data publishing requirements.

Every entry in the DGS database contains the type of the renewable energy power plant (photovoltaics, biomass, wind, hydropower, geothermal), the exact street address, the date of commissioning, and the power output capacity. The date of commissioning determines the applicable feed-in tariff (which, thereafter, is fixed for 20 years). For this reason, plant operators usually commission each system as soon as it is connected to the grid, as the feed-in tariff has been falling over time. We can thus exactly pinpoint when the installation of a PV system was finished. The power output capacity, in turn, is a measure of the size and, hence, the physical investment volume of a PV system. For PV systems, output capacity is measured by its peak energy production under ideal working conditions, denominated either in kilowatt peak (kWp) or megawatt peak (MWp, with 1 MWp equal to 1000 kWp).

We restrict attention to PV systems with an output capacity of less than 500 kWp that the DGS has not deemed to have errors such as wrong addresses.[9] This restriction is due to the fact that very large systems are usually mounted on the ground by few specialized companies, so that their installation does not affect local labor demand.[10] In addition, our instrumental variable is based on the net present value of rooftop systems and does not predict systems mounted on the ground.

From these raw data entries, we construct yearly physical investment in PV systems at the county level as the sum of installed capacity within a county in a given year. The main advantage of measuring physical investments, as opposed to monetary investments, is that physical investments capture real labor demand irrespective of variation in the prices of the production factors over time or across space. This is particularly relevant here, given that the price of solar panels varies considerably over time and given that the relevant wages may vary across space conditional on the state of the local labor market.

2.2. Determinants of rooftop PV installations

The volume of PV installations over time and across space is determined by five main factors: total costs, the feed-in tariff, the prevailing interest rate, solar radiation, and rooftop space. Total costs are obtained from an industry survey (Bundesverband Solarwirtschaft, 2012; 2014) that asks a representative sample of 100 companies about their total installation price per kWp. The resulting cost data are available quarterly since 2006. Prior to 2006, we use the yearly data from Janzing (2010). According to this data, the average installation costs of PV systems in our sample amount to € 3121 per kWp. We use this figure to calculate the costs of job creation based on the estimated employment effects of physical PV installations.

The second investment determinant is the feed-in tariff for electricity from photovoltaics specified by the Renewable Energy Act. The feed-in tariff typically varies by year; when there are multiple changes during a year (as in 2009 and 2012), we take the yearly average.[11]

The revenue flow from selling solar electricity at the price of the feed-in tariff accrues over time. The net present value of these revenues is calculated by discounting the expected payments using the "effective interest rates of commercial banks for housing loans," published by the Bundesbank.[12]

The profitability of PV systems is driven by their energy production, which is a function of the amount of solar radiation and the available rooftop space. Data on solar radiation is taken from the Photovoltaic Geographical Information System of the European Union (Huld et al., 2012). From the grid cell data on the "yearly average global irradiance on the optimally inclined surface," we calculate the average radiation (measured in kWh per square meter) at the county level.

For estimating the rooftop potential for solar energy production (in kWp), we follow the methodology of Lödl et al. (2010), who provide a detailed estimate of rooftop potential for the state of Bavaria. Lödl et al. (2010) first classify municipalities into four categories ("very rural," "rural," "suburban," and "urban") based on five observable municipality characteristics: population, population density, settlement area, average living area per capita, and the number of apartments per building. Second, they use aerial maps of 4500 dwellings to estimate the average rooftop potential conditional on the settlement area and the municipality's type. We apply their classification of municipalities to Germany and compute each municipality's rooftop potential using the conditional estimates from Lödl et al. (2010). Rooftop potential at the county level is given by the aggregate of these municipality-level estimates. Appendix A.2 provides a detailed description of the calculations.[13]

[9] A rule-of-thumb is that 1 kWp of power output capacity requires around 8–10 m^2 of space, implying that 500 kWp require around 4000–5000 m^2 (43,056–53,820 square feet) of space.

[10] This restriction affects fewer than 1% of PV systems installed between 2000 and 2012.

[11] The capacity bins that determine the exact feed-in tariff for each PV system have been changed in April 2012. Between April and December 2012, we use the feed-in tariff applicable to PV systems with a capacity of less than 10 kWp. Prior to April 2012, we use the feed-in tariff applicable to systems with less than 30 kWp capacity.

[12] This series starts in January 2003. For the calculation of the net present value in Panel (b) of Fig. 1, we use the series "average interest rates for mortgage loans" prior to 2003. This latter series has been discontinued in June 2003.

[13] A preferable approach would be to use building-level estimates of rooftop potential like Google's project "Sunroof." Google sunroof has been rolled out in Germany in May 2017, but only includes data for major municipalities thus far (https://www.eon-solar.de/, last accessed on March 12th 2018).

2.3. Employment data and control variables

The data on employment, unemployment, and wages is from the Federal Employment Agency, which collects this administrative data to determine social security contributions and eligibility. The employment data counts every employed individual who lives in a county and pays social security contributions, including part-time workers but excluding the self-employed and public servants. In our main analyses, we use the yearly mean of the quarterly available employment data measured on the last day of the quarter. The industry-specific employment data on the three-digit industry level is measured at the end of the second quarter and the monthly gross median wage in construction is measured at the end of the fourth quarter.

The covariates are either from the Federal Employment Agency or the Federal Statistical Office, unless noted otherwise. The data on county types ("non-city" or "city," where "city" is a county consisting of a single municipality) and on spatial planning regions are from the Federal Office for Building and Regional Planning.

In the empirical analyses, all variables are measured annually at the county level and normalized by a county's working-age population (between 15 and 65 years of age) in 2003 unless noted otherwise. This normalization facilitates the comparison of variables across counties. Appendix A.1 provides further details regarding the data.

3. Empirical model

The goal of our empirical strategy is to assess whether the effect of physical investments in rooftop PV systems on employment differs conditional on the state of the labor market. We identify the effect of investment in PV systems on employment by exploiting variation in installations within German counties from 2003 to 2012 using the following model

$$Employment\ p.c_{c,t} = \beta\ PV\ Installations\ p.c._{c,t} + CountyFE_c + \delta_{c,t}\ \mathbb{1}[Year_t \times State_c \times CountyType_c] + Controls_{c,t} + \varepsilon_{c,t}, \quad (1)$$

where the index c denotes the county, t denotes the year, and "p.c." (for "per capita") in the variable name indicates that the variable is normalized by the county's working-age population measured in 2003. PV installations are measured in megawatt peak (MWp).

We control for county fixed effects and year fixed effects for each county type and state combination (given by $\delta_{c,t}\ \mathbb{1}[Year_t \times State_c \times CountyType_c]$).[14] To adjust for labor market dynamics due to population flows, we control for population growth via the ratio of the working-age population in year t and the working-age population in 2003. We also account for construction activity as measured by the number of buildings completed in year t. Construction activity is likely to both affect the demand for rooftop PV installations and employment. We show in Appendix C that our results are unchanged for different sets of covariates. The standard errors are clustered at the level of 94 German spatial planning regions to account for potential geographic and serial correlation within these regions.[15]

In our main specification, we estimate Eq. (1) using OLS, separately for each sample of counties with tight and slack labor markets. In a second step, we repeat this analysis using the potential profitability of PV installations in a region as an instrumental variable for investment. The next section describes the identifying variation in PV installations across time and space as well as the construction of the instrument. Section 3.2. explains how we classify labor markets as being tight or slack.

3.1. Identifying variation

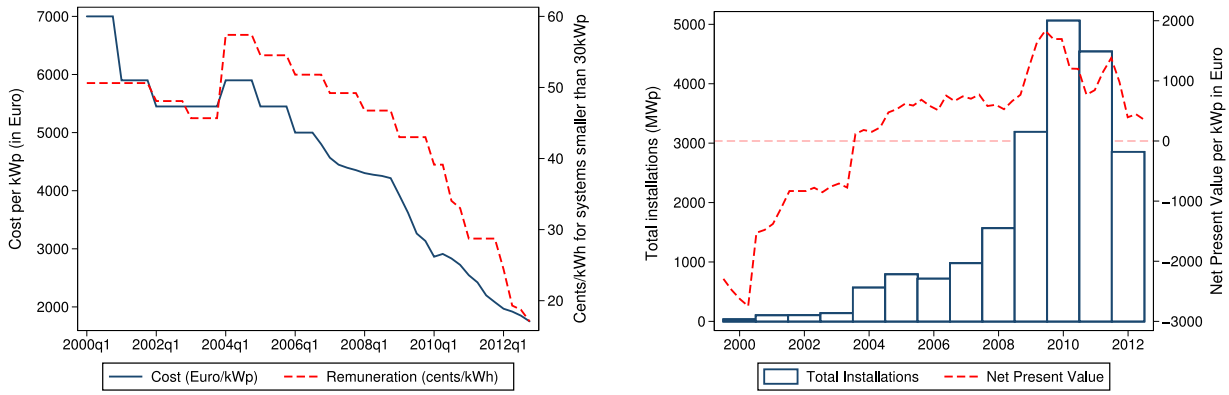
The profitability of a rooftop PV system is determined by how much electricity can be produced, how high this electricity is remunerated, and by the costs of installing and maintaining the system. The remuneration and costs exhibit substantial variation over time, but none across space, whereas the reverse is true for the potential for electricity production.

Determinants of investment over time The feed-in tariff is one of the time-varying determinants of the profitability of a rooftop PV-system. The dashed line in Panel (a) of Fig. 1 shows the feed-in tariff per kWh in Euro-Cents (right axis) of produced electricity for rooftop PV systems with less than 30 kWp, conditional on the date the system was connected to the grid. The initial feed-in tariff was 50.62 Cents in 2000, and scheduled to decrease by 5% each year from 2002 onwards. However, reflecting the policy goal of the government at the time to boost renewable energies, the feed-in tariff was raised to 57.40 Cents in 2004, with yearly degressions of 5% in 2005 and of 6.5% thereafter. The ensuing boom of solar energy production led to a steep increase in the cost of the policy. Further amendments of the law in 2009 and 2012 aimed to keep these costs in check, prescribing steeper degressions conditional on the volume of new installations in the previous year.

Falling costs of PV systems have also contributed to making PV installations more affordable, as illustrated by the solid line in Panel (a) of Fig. 1. Costs have declined steeply from € 7000 per kWp in 2000 to less than € 2000 per kWp in

[14] Counties are defined according to their boundaries in 2012, resulting in a total of 402 counties. We omit Hamburg and Berlin, as these are city-states and their employment outcomes are fully captured by the year fixed effects at the state level.

[15] Note that model (1) does not allow for dynamic effects of PV installations. Because PV installations are correlated over time, empirically separating their contemporaneous and lagged effects is difficult. Yet, there are reasons to believe that dynamic effects are small: first, the measure of installed solar power capacity captures only works that have been completed in a given year, and installing a PV system rarely takes longer than a few weeks. The direct employment effect should hence be confined to the same year. Second, Appendix B.1 follows an approach suggested by Shoag (2015) to provide evidence suggesting that dynamic effects are indeed negligible.



(a) Feed-In Tariff and Installation Costs (b) Total Installations and Net Present Value

Fig. 1. Feed-in tariff, costs, installations, and net present value of PV systems.
Note: Panel (a) shows the average costs (in Euro) per kWp for PV systems smaller than 100 kWp (solid line, left axis) as well as the feed-in tariff (fixed for 20 years, in Euro-Cents) per kWh of energy produced by a PV system smaller than 30 kWp conditional on when the system is connected to the grid (dashed line, right axis). Panel (b) displays the total installations of PV systems (measured in MWp) with less than 500 kWp capacity (bars, left axis) as well as the net present value (NPV) per kWp for a PV system with less than 30 kWp of capacity given the costs and the feed-in tariff from Panel (a) (dashed line, right axis). See the text for details.

2012. The drop in costs mainly reflects the global decline in the price of the system components (solar modules, power inverters). This decline has been caused by technological progress and higher competition due to the market entry of Asian manufacturers.

The increased feed-in tariff combined with rapidly falling costs made it profitable to invest into rooftop PV systems in most German regions. This is illustrated in Panel (b) of Fig. 1, which depicts the net present value for each kWp of credit-financed PV installations in a county with median solar radiation (dashed line) as well as total annual installations in MWp (bars).[16] In counties with median radiation and above, investing into rooftop PV systems became profitable with the increase in the feed-in tariff in 2004. The steep decline in costs in 2009 made the investment very profitable, particularly before lawmakers reacted by reducing the feed-in tariff accordingly. Yearly PV installations closely track the time variation in the profitability of these investments. In 2004, photovoltaic systems with 600 MWp were installed, more than in all previous years combined. After 2004, the upward trend continued until its peak in 2010, when yearly installations reached 5000 MWp.[17]

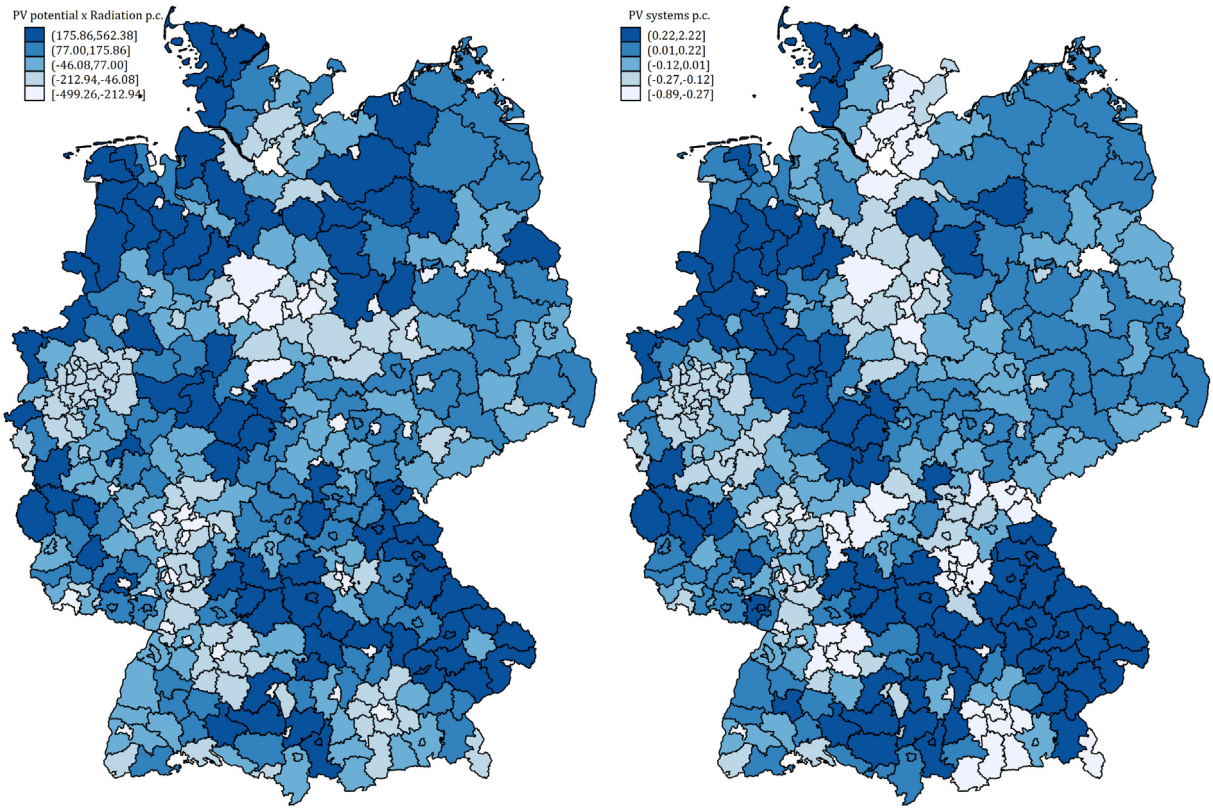
Determinants of investments across space As the feed-in tariff and the costs of PV systems are (roughly) equal across German regions, the extent to which counties may benefit from installing rooftop PV systems depends on the local potential for electricity production. The latter is, in turn, a function of the local amount of solar radiation and local rooftop potential, the space available and suitable for PV installations. Because the electricity produced by a PV system is proportional to the product of radiation and the amount of space covered with solar panels, a county's potential gain from PV installations is proportional to the product of the county's yearly radiation (measured in kWh per square meter) and its total rooftop potential (measured in kWp).

Panel (a) of Fig. 2 shows the spatial distribution of *rooftop potential* $\times$ *radiation*, normalized by the working-age population in 2003 and relative to the state-specific mean, across counties. While radiation is generally higher in the South, there is substantial variation in whether counties are more or less suitable for rooftop PV installations across all parts of Germany. Panel (b) of Fig. 2, in turn, depicts the spatial variation of total capacity installed during the major expansion of installations between 2003 and 2012, normalized by the working-age population and relative to the state-specific mean, as before. Comparing the variation in *rooftop potential* $\times$ *radiation* and PV installations, it becomes clear that counties with a higher suitability for PV installations in general also experience a larger increase in their solar power capacity.

Remuneration potential The time-varying costs and benefits of installing PV systems can be combined with the regional productivity of PV systems in producing solar energy into a single measure that captures the time-variation in local profitability of PV installations. This measure, which we call "*remuneration potential*" and which we use as an instrument for investments in Section 4.2, is the net present value of investing in PV systems with an output capacity equal to the county's

[16] Most rooftop PV systems are, at least in part, credit financed (Bickel et al., various years). We assume that the relevant interest rate for financing PV installations is similar to the one for mortgages, as the PV system and its relatively risk-free income stream can serve as collateral. The average yearly interest rate on new mortgages has fluctuated between 4% and 6.5% between 2000 and 2009, and has been dropping to below 3% between 2009 and 2012. Eq. (2) below gives the exact formula for the net present value.

[17] For comparison, one reactor of a typical nuclear power plant produces between 500 MW and 1500 MW of electrical power.



(a) Radiation × Rooftop Potential p.c. (b) Total Installations p.c.

Fig. 2. Geographic distribution of total installations and rooftop potential × radiation.
Note: Panel (a) shows the geographic distribution of *rooftop potential* × *radiation* per capita (p.c.) across counties relative to their state-specific mean. Panel (b) depicts the*total power output capacity* (in kWp p.c.) installed across counties between 2003 and 2012 relative to the state-specific mean. The city-states of Hamburg (blank county in the North) and Berlin (blank county in the North-East) are excluded. The color coding scheme corresponds to quintiles of installations and *rooftop potential* × *radiation*; darker colors indicate higher values. Per capita values are normalized with the working-age population in 2003.

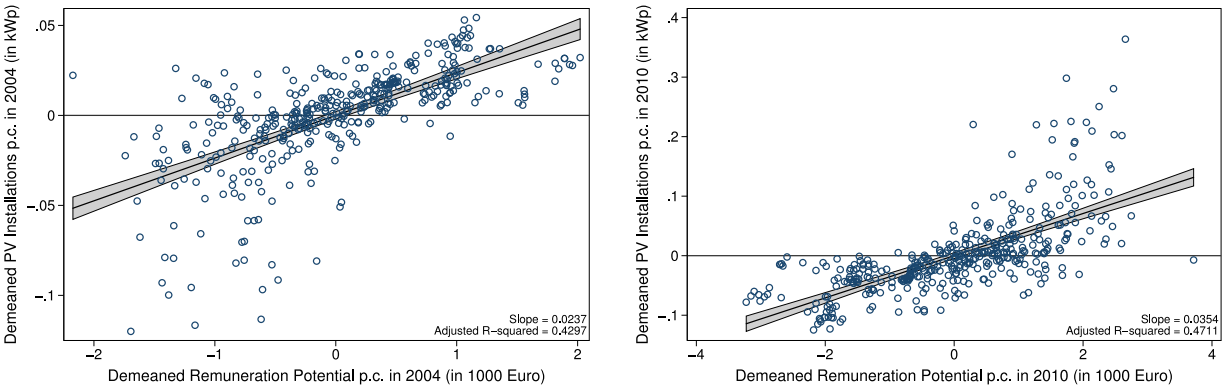
rooftop potential in a given year t. Formally, the remuneration potential of county c in year t is defined as follows:

$$Remuneration\ Potential_{c,t} = Rooftop\,Potential_c \times \left[\sum_{\tau=t}^{t+20} \left(\left(\frac{1}{1+i_t} \right)^{\tau-t} \left(\underbrace{0.995^{\tau-t} \cdot 0.75 \cdot Radiation_c}_{\text{electricity produced by 1 kWp system}} \cdot Tariff_t - \underbrace{0.01 \cdot Costs_t}_{\text{operating costs}} \right) \right) - Costs_t \right]. \tag{2}$$

Remuneration potential is the product of the rooftop space suitable for PV systems, measured in kWp, and the net present value of operating a PV system with output capacity of 1 kWp for 20 years from year t onwards.[18] The net present value, the term in brackets, is given by the net income stream (discounted using the interest rate at t, i_t) less the installation costs at t. The net income stream, in turn, equals the electricity production times the feed-in tariff, where we need to adjust the power output capacity under optimal conditions for average working conditions. Here, we follow the European Union's PVGIS and assume a performance ratio of 0.75. Following Wirth (2015), we also adjust for gradual performance losses of 0.5% per year and annual operating costs of 1% of the installation costs.

Fig. 3 shows that remuneration potential is a strong predictor of investments in PV systems.[19] It plots PV installations per capita, demeaned by their 2003–2012 county mean and relative to the state × year average, against the similarly demeaned remuneration potential per capita. Panel (a) shows the data for the start of the PV investment boom in 2004, and Panel (b) shows the data for the peak of the boom in 2010. In both years, variation in remuneration potential explains roughly half of the variation in PV installations. The slopes imply that a € 1000 increase in per capita remuneration potential is associated

[18] We assume the PV system to be operational for 20 years, as this is the time for which the feed-in tariff remains fixed.
[19] This echoes the formal first stage results in Appendix B.2.



(a) In 2004 (b) In 2010

Fig. 3. Remuneration potential and PV installations.
Note: This figure plots demeaned PV installations per capita (relative to the county and state × year specific mean) against the identically demeaned remuneration potential per capita as defined by Eq. (2). Panel (a) plots these data for the cross-section of counties in 2004, and Panel (b) shows the equivalent data for 2010.

with an increase in per capita installations of 0.024–0.035 kWp. Given the installation costs in 2004 and 2010, the latter correspond to additional investments worth roughly € 140 and € 100, respectively.

3.2. Classification of slack and tight labor markets

In order to investigate whether the employment effect of PV installations depends on the state of the economy, we need to define whether a labor market is "slack" or "tight." We build on the approaches of Nakamura and Steinsson (2014) and Shoag (2015). In particular, we define the state of the labor market of a county at time t as slack if the county's unemployment at time $t-1$ is above a benchmark. Otherwise, the county's labor market is defined as tight. Given this definition, the choice of the benchmark specifies in which dimension the sample is split into counties with slack and tight labor markets. For each sample split, we then estimate Eq. (1) separately for each subsample and test whether the employment effects of investments differ between the subsamples.

For the main specifications, we apply two definitions of the benchmark unemployment level that separates slack and tight labor markets. The first definition follows Nakamura and Steinsson (2014) and splits the sample along the time series dimension. This exploits the long, ten year period, during which there were stable and favorable conditions for investments in PV systems. According to this *time-series split*, a county's labor market is defined to be slack if its unemployment in the previous year is higher than the county's mean unemployment between 2003 and 2012.[20] Panel (a) of Fig. 4 shows that, according to the time-series split, most counties' labor markets are defined to be slack prior to 2008 and tight thereafter. This reflects the downward sloping trend in German unemployment over this period.

The second definition of the unemployment benchmark follows Shoag (2015) and splits the sample along the cross-sectional dimension. According to this *cross-sectional split*, a county's labor market is defined to be slack if its unemployment in the previous year is higher than the state mean of unemployment in the same year.[21] Panel (b) of Fig. 4 shows that the cross-sectional split selects a similar set of counties to have slack and tight labor markets in the different years.

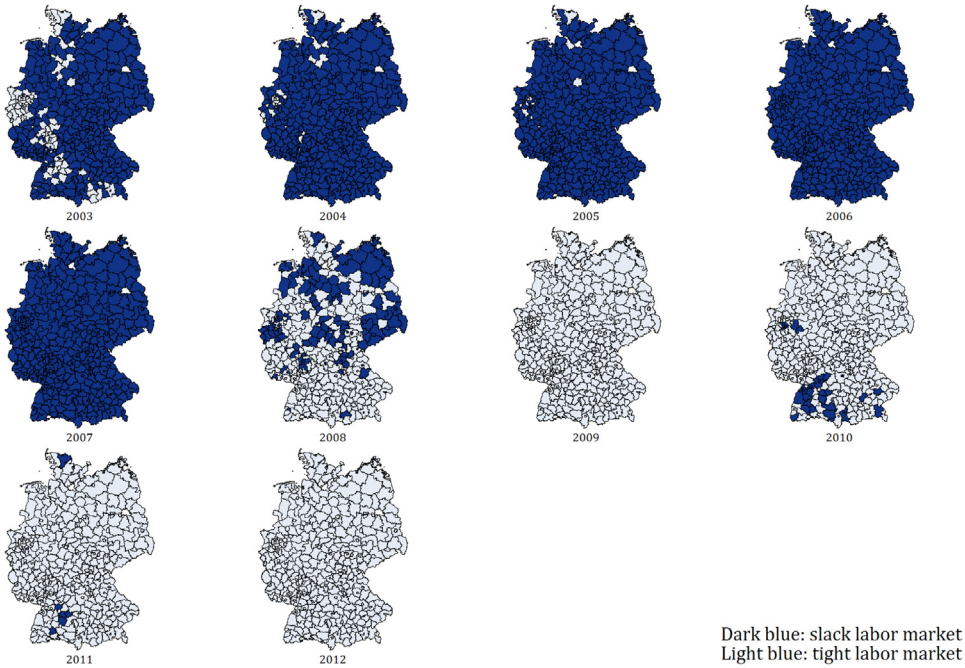
The two sample splits have different implications. The time-series split compares the same set of counties at different times, so that the two samples of slack and tight labor markets, respectively, share the same structural features. The cross-sectional split compares different counties at the same time, thus holding constant all factors that may affect the employment response to investments over time (such as innovation in the production technology). Hence, differential employment effects in the sample splits along both dimensions can neither be explained by time trends nor structural features alone. To further bolster the robustness of the results with respect to the definition of slack and tight labor markets, Section 4.3 explores the employment response to investments for a wide range of additional definitions.

4. Results

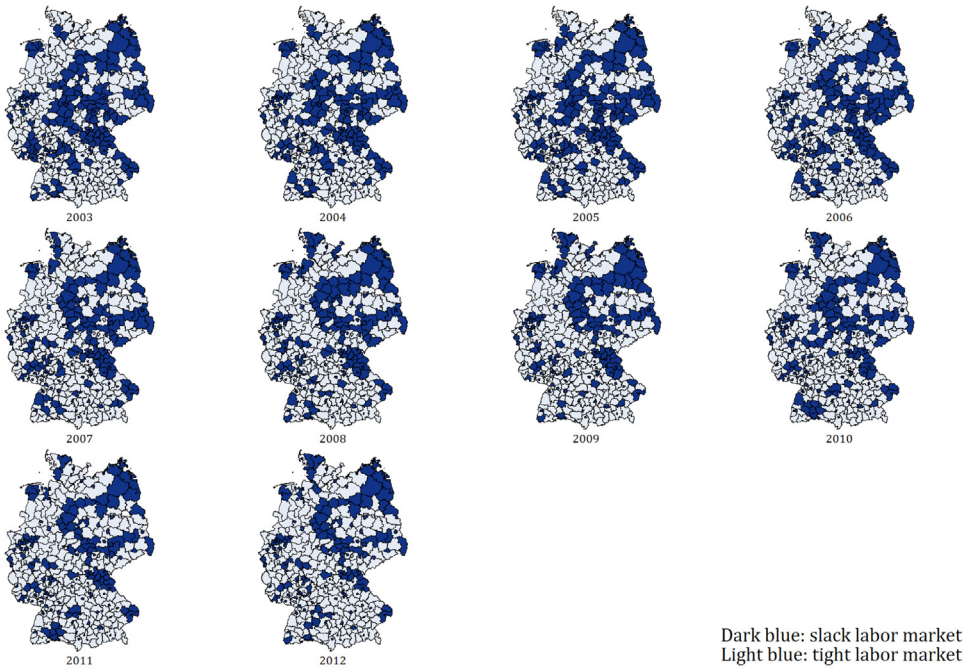
This section presents the main findings. The empirical analysis shows that physical investments increase employment more at times and in regions with slack labor markets compared to times and regions with tight labor markets. This result

[20] Formally, the labor market is said to be slack according to the time-series split if $unemployment_{c,t-1} > 1/10 \sum_{t=2003}^{2012} unemployment_{c,t}$.

[21] Formally, the labor market is said to be slack according to the cross-sectional split if $unemployment_{c,t-1}/N_{c,2003} > (\sum_{c \in state(c)} unemployment_{c,t-1})/(\sum_{c \in state(c)} N_{c,2003})$, where $N_{c,2003}$ is county c's working-age population in 2003.



(a) Time-series Classification of Slack



(b) Cross-sectional Classification of Slack

Fig. 4. Labor market slack across counties.
Note: Panel (a) shows, for each year in the sample, which counties are classified as having a slack or tight labor market according to the definition of slack in the time-series dimension. Here, the labor market is defined to be slack if a county's unemployment in the previous year is higher than the mean unemployment of the county over the sample period. Panel (b) shows which counties exhibit slack / tight labor markets according to the definition of slack in the cross-sectional dimension. Here, the labor market is defined to be slack if a county's unemployment in the previous year is higher than the state average of unemployment in the previous year.

Table 1
The effect of PV installations on employment (OLS).

Split along	Employment rate				
		Time series		Cross-section	
	Baseline (1)	Slack (2)	Tight (3)	Slack (4)	Tight (5)
Installed capacity p.c. (MWp)	19.98***	36.61**	2.78	37.91***	13.34**
	(6.58)	(17.09)	(3.86)	(13.83)	(6.47)
Population growth	0.37***	0.37***	0.25***	0.31***	0.40***
	(0.02)	(0.03)	(0.02)	(0.04)	(0.03)
Construction p.c.	−0.25	−0.19	0.47	0.04	−0.23
	(0.22)	(0.14)	(0.30)	(0.21)	(0.26)
County fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
P-value slack < tight		0.017		0.036	
Jobs per € 100,000	0.64	1.17	0.09	1.21	0.43
Observations	4000	2044	1956	1783	2189

Note: The dependent variable is the average yearly employment rate (employment normalized by the working-age population in 2003) between 2003 and 2012. *Installed capacity p.c.* are yearly PV installations measured in megawatt peak (MWp) normalized by the working-age population in 2003 (indicated by "p.c." for "per capita"). *Population Growth* is the ratio of the working-age population in a given year to the working-age population in 2003. *Construction p.c.* is the number of residential and non-residential buildings completed in a given year. *P-value slack < tight* reports the *p*-value of the test of the null hypothesis that the employment effect of PV installations is smaller in slack than in tight labor markets. The *year fixed effects* are estimated at the level of the state × county type (rural or urban county). Note that the observations in the cross-sectional split do not sum to 4000 due to singleton groups. Standard errors (in parentheses) are clustered at the level of 94 spatial planning regions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

holds irrespective of whether we estimate the employment effects of investments via OLS or IV, and independent of the particular definition of slack and tight labor markets.

4.1. OLS results

Table 1 presents the OLS estimates of empirical model (1). As a benchmark, column (1) reports the average effect of physical investments in PV systems on employment for the full sample. Because both employment and installations are normalized by the working-age population, the coefficient of *installed capacity p.c.* can be interpreted as the number of additional jobs per MWp of PV installations. Hence, in the full sample, additional PV installations of 1 MWp capacity lead to, on average, around 20 additional jobs lasting for one year. Given the average installation costs of € 3.121 million per 1 MWp capacity, this estimate implies that investments of € 100,000 created 0.64 local job-years, corresponding to costs per job-year of € 156,000.

The magnitude of this estimate is in line with how much work is required to install rooftop PV systems, the direct effect of additional demand on the labor market: The industry survey of Brohm (2010) suggests that, in 2008, installing capacity of 1 MWp required on average between 15 and 20 workers. We also have access to the internal accounting data of one installation company. Their data suggests a lower bound of 9 worker-years per MWp.[22]

Columns (2) and (3) report the estimates for the sample split into slack and tight labor markets along the *time series* dimension. Here, we find that at times of economic slack, additional PV installations of 1 MWp capacity lead to about 37 additional job-years, corresponding to 1.17 job-years per investments of € 100,000. This effect is 80% larger than the baseline effect in column (1). In contrast, at times of tight labor markets, the local employment effect of investments into PV systems is economically small and statistically indistinguishable from zero. Moreover, we reject the null hypothesis that the investment-induced employment gains are larger at times of tight labor markets at the five percent level; the *p*-value of the respective one-sided test equals 0.017.

The *cross-sectional* split in columns (4) and (5) leads to similar results as the time series split. Here, 1 MWp in PV installations leads to 38 more job-years in counties with slack labor markets, while the same additional demand creates only 13 new job-years in counties with tight labor markets. As before, the null hypothesis that employment gains in regions with tight labor markets are larger than in ones with slack labor markets is rejected at the five percent level.

The differences in the employment effects across the two sample splits cannot be explained by either counties' structural characteristics or time-varying changes in the relation between real demand and employment alone. While a correlation of local investments and structural labor market characteristics–such as higher investments in less sklerotic labor markets–may explain the difference in the employment effects in the cross-sectional split in columns (4) and (5), such an explanation cannot account for the difference in the employment creation over time in the identical sets of counties in the time series split. Conversely, we have seen in Panel (a) of Fig. 4 that local labor markets were mostly slack in the first years of the

[22] This number includes the installation of the panels, the acquisition and the planning of the PV system as well as the additional jobs at the regional distributor. It does not include jobs outside of the installation industry. The 9 worker-years are within the 95% confidence interval of our OLS estimate in column (1). If we only consider employment in the high exposure sector-as we do in Table D.1 in Appendix D.2-we arrive at 15.04 job-years per MWp with a standard error of 4.74. Here, 9 job-years are within the 80% confidence interval.

photovoltaic investment boom and tight in the later years, so that the results in columns (2) and (3) could potentially be explained by a reduction in labor demand for new installations.[23] However, changes in technology cannot account for the results of the cross-sectional split, as the latter compares different counties with slack and tight labor markets at the same time. Additionally, the next section uses an instrumental variable approach to provide additional evidence that unobserved third factors are unlikely to drive the results of Table 1.

To interpret the magnitude of the employment effect, we approximate the local labor earnings multiplier. To do so, we divide the median annual earnings by the costs per job-year implied by our estimates. Median earnings equaled € 32,160 during the sample period. The results in column (1) of Table 1 imply costs per job-year equal to € 100,000/0.64 = €156,000. Combined, this translates into a local earnings multiplier of 0.21, i.e., local earnings increased by € 0.21 per € 1 investment in PV systems. In slack labor markets, this multiplier is around 0.38, while in tight labor markets the multiplier is at most 0.14. Finally, it is important to note that only around a third of the total costs of PV systems, the basis for the calculation of the costs per job-year, accrue locally.[24] If we scale our estimates accordingly, the multiplier in tight labor markets is around 1.14, while in slack labor markets the multiplier is at most 0.42.

4.2. IV results

One concern for identifying the effect of PV installations on employment is that investment decisions may depend on expected labor market dynamics via an unobserved third channel, such as local credit markets. As many rooftop PV systems are credit-financed, changes in local lending might influence both, employment and investments and thus bias the estimates in either direction.[25] For example, OLS could overestimate the effect of PV installations on employment if favorable lending conditions drive employment growth and investment. OLS could also underestimate the effect of PV installations on employment if loans for safe investments into PV systems are particularly attractive when the local economy is on a downward trajectory.

To address such concerns, we instrument local investments into PV systems by their profitability which is captured by *remuneration potential* as defined in Section 3.1. *Remuneration potential* is a function of time-varying factors (the feed-in tariff, the costs of components, mortgage rates) that are determined at the global or national level and thus unrelated to the trajectories of local labor markets, as well as by pre-determined geographic characteristics (the rooftop potential, local solar radiation) that are likely fixed over time and thus unresponsive to labor market developments as well. At the same time, *remuneration potential* strongly predicts investments, as shown in Fig. 3. Taken together, the variable *remuneration potential* hence likely meets the identifying assumptions of relevance and exogeneity. Appendix B.2 provides additional details regarding the first stage and further arguments for why the exclusion restriction is likely to hold.

Table 2 summarizes the IV estimates of the main empirical model (1). The IV results are qualitatively similar to the findings from OLS. The employment gains due to PV installations in slack labor markets are larger than the overall average and much larger than the employment gains in tight labor markets. At average installation costs, the estimated employment gains from physical investments imply that € 100,000 in PV installations increase employment by about five job-years in slack and by about one job-year in tight labor markets, both in the time series and the cross-sectional split.[26] Due to these large differences, we reject the hypothesis that employment is more responsive to investments in tight than in slack labor markets at the one percent level, even though the IV estimates are less precise than the corresponding OLS results. Finally, note that with Kleibergen-Paap F-statistics of the excluded instruments equal to 22 or higher, the first stage is strong in all specifications.

4.3. Alternative classifications of slack and tight labor markets

Table 3 uses alternative classifications of slack and tight labor markets to show that our finding of differential employment effects does not crucially depend on the specific classification.[27] Each row of Table 3 reports the results of an alternative sample split. Panel A reports the OLS estimates and Panel B the corresponding IV results.

[23] Contrary to this hypothesis, conversations with industry experts suggest that there was no fundamental change to the technology for installing PV systems over time.

[24] According to an industry survey in 2013, the local installation costs amount to about 20% of the total costs, while the remaining 80% are spent on solar panels and components (EuPD Research, 2013). According to anecdotal evidence from industry experts, installation firms charge an additional 10% of the total costs as a mark-up on the panels and components, so that roughly one third of the total costs contribute to local demand.

[25] Financial service provision in Germany has a strong regional focus due to the nationwide presence of local savings banks (*Sparkassen*) and credit cooperations (*Volks- und Raiffeisenbanken*). In 2012, there were 423 savings banks, the area of business of which is often defined by county borders, and more than 900 credit cooperations. Statistics on the share of debt-financing of PV systems do not exist. However, the state-owned bank *Kreditanstalt für Wiederaufbau (KfW)* reports that in the years 2007–2012, between 42% and 74% of the yearly investments in PV systems have been at least partially backed by their subsidized loans program (Bickel et al., various years).

[26] The difference in the magnitude of the OLS and IV estimates could be driven by higher investment incentives in less prosperous regions that lead to a downward bias of the OLS estimates. Another part of the explanation may be that the IV estimates are, despite their strong first stage, much more noisy than the OLS estimates. Furthermore, the IV estimates a local average treatment effect, while the OLS uses the entire variation in the data. Local average treatment effects are difficult to interpret because the compliers, for which the treatment effects are estimated, cannot be identified in the data. We thus cannot assess whether the IV results can be extrapolated to the entire population.

[27] In addition, in Appendix C.3 we show that for the main slack definitions, the results are comparable when we split observations into three groups instead of two.

Table 2
The Effect of PV Installations on Employment (IV).

Split along	Employment rate				
		Time series		Cross-section	
	Baseline (1)	Slack (2)	Tight (3)	Slack (4)	Tight (5)
Installed capacity p.c. (MWp)	52.57***	148.43***	22.60**	180.11***	30.52**
	(13.60)	(45.38)	(10.86)	(37.73)	(12.01)
Population growth	0.36***	0.36***	0.25***	0.30***	0.38***
	(0.02)	(0.03)	(0.02)	(0.03)	(0.03)
Construction p.c.	−0.24	−0.20	0.48*	0.36	−0.25
	(0.23)	(0.14)	(0.29)	(0.30)	(0.27)
County fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
P-value slack < tight		0.003		0.000	
Jobs per € 100,000	1.68	4.76	0.72	5.77	0.98
F-statistic instrument	88.22	22.11	59.15	28.43	79.67
Observations	4000	2044	1956	1783	2189

Note: The dependent variable is the average yearly employment rate (employment normalized by the working-age population in 2003) between 2003 and 2012. *Installed capacity p.c.* are yearly PV installations measured in megawatt peak (MWp) normalized by the working-age population in 2003 (indicated by "p.c." for "per capita"). *Installed capacity p.c.* is instrumented by *remuneration potential p.c.* as defined in Section 3.1. *Population growth* is the ratio of the working-age population in a given year to the working-age population in 2003. *Construction p.c.* is the number of residential and non-residential buildings completed in a given year. *P-value slack < tight* reports the *p*-value of the test of the null hypothesis that the employment effect of PV installations is smaller in slack than in tight labor markets. *F-statistic instrument* is the Kleibergen-Paap F-statistic of *remuneration potential p.c.* in the first stage. The *year fixed effects* are estimated at the level of the state × county type (rural or urban county). Note that the observations in the cross-sectional split do not sum to 4000 due to singleton groups. Standard errors (in parentheses) are clustered at the level of 94 spatial planning regions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Rows (1) and (8) in boldface repeat the baseline time series and cross-sectional splits from Tables 1 and 2, respectively, for comparison purposes. Row (2) contains an alternative time series split that defines all years prior to 2007 as times of slack and all years after 2008 as times of tight labor markets. Rows (3) to (6) split the sample based on unemployment benchmarks calculated across time and space. In row (3), a county is said to have a slack labor market if its unemployment rate in year $t-1$ is above the average national unemployment rate between 2003 and 2012. Otherwise, the labor market is said to be tight. In rows (4) and (5), labor markets are classified accordingly, but with respect to the 2003–2012 state average and the 2003–2012 state × county-type average, respectively.[28] In row (6), we regress unemployment on county and state-year fixed effects and split the residuals from this regression at zero to determine whether a county's labor market is classified as slack or tight. Positive residuals indicate a slack market and negative residuals indicate a tight market. Rows (7) to (10) provide cross-sectional splits. In row (7), a labor market is defined as slack if its unemployment rate in $t-1$ is above the national mean in $t-1$. Slack and tight labor markets in rows (8) and (9) are defined similarly, but with respect to the state and state × county-type average in $t-1$, respectively. In row (10), a labor market is said to be slack if in 2002, the last year before the sample period, its unemployment rate was above the 2002 state mean, and said to be tight otherwise. Finally, in row (11) we use the year-over-year growth of median wages in the construction sector as a measure of slack. If the wage growth in a county lags behind national inflation in $t-1$, the labor market is defined as slack and tight otherwise.[29]

The estimated employment effects of investments in slack labor markets are above the corresponding estimates for tight labor markets in all specifications of Table 3. Except in the OLS specifications in row (9) and the IV specification in row (6), these differences are economically meaningful with the employment gains in slack market conditions being at least 50 percent larger than the ones in tight conditions. The coefficients are statistically different from each other on conventional levels in all but four of the 22 specifications.

Taken together, we consistently find that more jobs are created in slack than in tight markets. This empirical pattern is present irrespective of the exact definition of labor market slack and robust to using an IV strategy instead of OLS.[30]

5. Discussion of mechanisms

There are several potential explanations for the empirical pattern of fewer jobs being created locally when the labor market is tight. First, there may be a larger incentive to substitute labor with capital. Second, installation firms might meet

[28] This means that in row (4), a county is said to have a slack labor market if its unemployment rate in $t-1$ is above the average unemployment rate between 2003 and 2012 in the same state. In row (5), the unemployment rate in $t-1$ is compared to the average unemployment rate between 2003 and 2012 within all counties of the same county type (either urban or rural) and the same state.

[29] For this definition of labor market tightness the sample size is smaller than for the other specifications as the wage data is only available for counties with more than 1000 employees in construction due to data confidentiality requirements.

[30] Appendices C.1 and C.2 provide further robustness checks regarding data and covariate choices.

Table 3
Alternative definitions of slack in the labor market.

	Slack		Tight		*P*-Value
	Coeff (1)	SE (2)	Coeff (3)	SE (4)	(5)
Panel A: OLS					
Based on time variation in unemployment					
(1) **Within county mean**	148.43	45.38	22.60	10.86	0.003
(2) 2003–2007 vs. 2008–2012	200.21	49.30	11.45	6.44	0.000
Based on aggregate variation in unemployment					
(3) 2003–2012 national mean	28.33	26.35	18.69	6.93	0.361
(4) 2003–2012 state mean	49.82	18.22	11.25	6.47	0.013
(5) 2003–2012 state × county type mean	32.09	9.38	14.16	6.99	0.049
(6) Residuals wrt. county and state × year FEs	20.46	5.60	12.50	6.01	0.058
Based on cross-sectional variation in unemployment					
(7) Yearly national mean	43.04	23.60	21.22	7.30	0.186
(8) **Yearly state mean**	37.91	13.83	13.34	6.47	0.036
(9) Yearly state × county type mean	23.56	7.58	19.61	7.17	0.296
(10) State mean in 2002	31.32	12.00	12.78	6.31	0.061
Based on cross-sectional variation in construction wage growth					
(11) National CPI growth	22.59	7.12	13.10	6.62	0.038
Panel B: IV					
Based on time variation in unemployment					
(1) **Within county mean**	148.43	45.38	22.60	10.86	0.003
(2) 2003–2007 vs. 2008–2012	200.21	49.30	11.45	6.44	0.000
Based on aggregate variation in unemployment					
(3) 2003–2012 national mean	142.24	73.87	32.50	12.00	0.071
(4) 2003–2012 state mean	193.67	48.04	24.54	10.68	0.000
(5) 2003–2012 state × county type mean	153.87	35.26	18.44	11.38	0.000
(6) Residuals wrt. county and state × year FEs	59.97	15.48	45.32	14.85	0.167
Based on cross-sectional variation in unemployment					
(7) Yearly national mean	131.08	53.86	44.45	14.32	0.057
(8) **Yearly state mean**	180.11	37.73	30.52	12.01	0.000
(9) Yearly state × county type mean	98.36	24.63	33.41	13.37	0.006
(10) State mean in 2002	120.52	31.05	31.98	12.58	0.002
Based on cross-sectional variation in construction wage growth					
(11) National CPI growth	59.86	13.28	30.38	12.90	0.005

Note: This table presents the employment effects of PV installations in slack and tight labor markets for various alternative sample splits. Except for rows (2), (6), (10) and (11), the name of each row specifies a different unemployment benchmark (e.g., in row (3) the benchmark is the 2003–2012 national unemployment mean). The labor market is said to be slack if unemployment in $t-1$ is above the benchmark, and said to be tight otherwise. In row (2), labor markets are defined as being slack in 2007 and earlier, and tight in 2008 and later. In row (6), a labor market is slack if the residual of regressing unemployment on county and state-year fixed effects is positive in $t-1$. In row (10), a labor market is slack if its unemployment rate in 2002 was higher than the state mean in 2002, and tight otherwise. In row (11), a labor market is slack if its year-over-year growth of the median wage in construction is lower than national inflation. Panel A reports the OLS results, and Panel B reports the IV results with the same model specification as in Tables 1 and 2, respectively. Columns entitled "Coeff" report the OLS/IV coefficient estimate of *installed capacity p.c.* for the subsample with slack and tight labor market, respectively. Columns entitled "SE" report the corresponding standard errors, clustered at the level of 94 spatial planning regions. The column entitled "*P*-Value" report the *p*-values of the test of the null hypothesis that the employment effect of PV installations is smaller in a slack labor market than in a tight labor market.

their labor demand with hiring workers from outside the local labor market. Third, investment in tight labor markets might lead to crowding out, as in Michaillat (2014). Workers installing PV systems might be effectively drawn from other jobs if the labor market is tight-putting pressure on wages in the process, while they might have been unemployed if the labor market is slack. Finally, there may be direct effects of labor market tightness on the costs of investment (e.g., due to higher wages for installation workers or higher markups) implying a lower effective labor demand for each Euro invested in a tight labor market. However, the latter effect cannot play a role here, as we use the physical investment volume as explanatory variable.[31]

Effects across industries The first channel, a substitution of labor with capital in tight labor markets, is unlikely to drive the results in the context of PV investment. The production function of installing PV systems has been stable over the entire time period. The production process consists of bringing the PV systems to the customer and workers carrying solar panels onto rooftops and mounting them there, with little scope for different installation techniques. The only substitute technology available for installation are telehandlers that can lift the panels onto roofs, which still require extensive manual labor to mount and install the system. Unfortunately, we do not have data on the usage of telehandlers and hence we

[31] Another potential explanation for differential effects in tight and slack markets are nonlinearities in the response of employment to PV installations. To allow for non-linear effects, Appendix D.1 computes the marginal effects from a variant of model (1) that adds the installed capacity squared as a covariate. This analysis shows that nonlinearities cannot account for the differential employment creation in slack and tight markets.

Table 4
Sectoral employment conditional on slack: OLS results.

	Industry-specific employment p.c.					
	High exposure		Local		Other	
	Slack (1)	Tight (2)	Slack (3)	Tight (4)	Slack (5)	Tight (6)
Panel A: Time series split						
Capacity p.c. (MWp)	17.96***	2.98**	7.67**	3.94***	20.97	-4.39
	(6.49)	(1.37)	(3.87)	(1.23)	(23.76)	(3.29)
P-value slack < tight	0.012		0.162		0.136	
Jobs per € 100,000	0.58	0.10	0.25	0.13	0.67	-0.14
Observations	2044	1956	2044	1956	2044	1956
Panel B: Cross-sectional split						
Capacity p.c. (MWp)	17.10**	11.45***	15.18***	6.64***	5.94	-5.45
	(8.37)	(4.17)	(3.16)	(2.54)	(15.94)	(8.47)
P-value slack < tight	0.220		0.008		0.211	
Jobs per € 100,000	0.55	0.37	0.49	0.21	0.19	−0.17
Observations	1783	2189	1783	2189	1783	2189
PopGrowth & construction	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable in columns (1) and (2) is employment in the high exposure sectors normalized by the working-age population in 2003 (indicated by "p.c." for "per capita"). The dependent variable in columns (3) and (4) is employment p.c. in local industries. The dependent variable in columns (5) and (6) is employment p.c. in all remaining industries. Employment by industry is measured annually on June 30th. Table A.1 in Appendix A.1 provides details of the industry classifications. *Capacity p.c.* are yearly PV installations measured in megawatt peak (MWp). Except for the dependent variables, the empirical specifications are identical to the ones in Table 1. In particular, the *year fixed effects* are estimated at the level of the state × county type (rural or urban county). Panel A reports the results for the time series split, and Panel B reports the results for the cross-sectional split. Standard errors (in parentheses) are clustered at the level of 94 spatial planning regions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

provide indirect evidence that the usage of telehandlers is unlikely to account for the differences in employment gains in slack and tight labor markets.

To this end, we partition total employment into employment in (i) high exposure industries, (ii) local non-tradables, and (iii) all other industries. The high exposure industries are directly affected by the demand for PV installations, such as electricians.[32] We classify the retail and wholesale sector, the hospitality industry (hotels and restaurants) as well as financial service providers as local services. Companies in these industries might benefit from local demand spillovers. All remaining industries are classified as belonging to "other industries."[33] If it were the case that the difference in slack versus tight labor markets was driven by changes in the installation technology, we would expect that the differential response is entirely driven by differences in employment gains in the high exposure industries. Employment creation in local industries should not exhibit differential effects, as the labor-saving technology is specific to the installation of PV systems.[34]

Table 4 reports the OLS estimates of the employment gains due to PV installations in slack and tight labor markets for each of the three sectors. Panel A presents the results for the time series split, and Panel B presents the results for the cross-sectional definition of labor market slackness. The first finding from Table 4 is that in both slack and tight labor markets, PV installations led to a statistically significant increase in employment only in the high exposure and local sectors. In contrast, the employment gains (or losses) in all other sectors are very imprecisely estimated. Given the nature of the investments we study, this is exactly the pattern of employment gains across industries we expect to find.[35]

A second result from Table 4 is that the difference of the employment gains in slack and tight labor markets is driven by differential employment gains both in high exposure and local industries. In the time series as well as the cross-sectional split, the difference in the employment gains between slack and tight markets is sizable, and, in two cases, significantly different from each other.[36] This speaks against adjustments in the production technology of PV installations conditional on the state of the labor market as an explanation for the differential employment gains in slack and tight labor markets.

[32] For classifying industries as being "high exposure," we take a random sample of firms that are a member of the *Bundesverband Solarwirtschaft* (a trade association of the German solar industry) that install PV systems and consult their Creditreform company profiles to extract their industry classification. Most of the sampled firms are certified electricians; as such, they belong to various industries, including building installation and engineering. The union of the industry codes identified by this procedure constitutes the set of high exposure industries.

[33] "Social services" (industry code 853) is excluded from "other industries" for two reasons. First, the employment data in this industry is non-stationary, as it increases from 4.4% to 6.2% of the workforce from 2003 to 2012. Second, this sector mostly comprises of the daycare industry (care of elderly and children), and it is unclear whether these are local services or, indeed, "other industries." Table A.1 in Appendix A.1 lists the assignment of industry codes to each of the three subsectors.

[34] Note that this additionally assumes that the spillovers from the high exposure to the local sectors do not change with the state of the labor market.

[35] This mirrors the result for the full sample of counties in Table D.1 in Appendix D.2, where we find that around 60% of the entire employment effect originates from the high exposure sectors, while 40% originates from local industries.

[36] The corresponding IV results reported in Table D.3 in Appendix D.3 mirror the OLS findings.

Table 5
Spillovers from neighboring counties: OLS results.

	Employment rate					
	Planning region		5 closest counties		10 closest counties	
	Slack (1)	Tight (2)	Slack (3)	Tight (4)	Slack (5)	Tight (6)
Panel A: Time series split						
Capacity p.c. (MWp)	32.58**	2.09	34.28**	2.81	32.80**	1.76
	(16.39)	(4.04)	(15.79)	(4.38)	(16.23)	(4.29)
Neighboring capacity p.c.	3.05	0.43	0.72	−0.01	0.91	0.23
	(2.66)	(0.75)	(1.67)	(0.74)	(0.69)	(0.39)
P-value slack < tight	0.024		0.017		0.020	
Jobs per € 100,000	1.04	0.07	1.10	0.09	1.05	0.06
Observations	2044	1956	2044	1956	2044	1956
Panel B: Cross-sectional split						
Capacity p.c. (MWp)	42.25***	13.46**	41.53***	15.36***	37.98***	15.56***
	(11.20)	(6.54)	(12.18)	(5.84)	(10.28)	(5.90)
Neighboring capacity p.c.	−0.86	−0.10	−0.67	−0.86	−0.01	−0.59
	(0.93)	(2.06)	(0.88)	(1.33)	(0.57)	(0.64)
P-value slack < tight	0.010		0.022		0.024	
Jobs per € 100,000	1.35	0.43	1.33	0.49	1.22	0.50
Observations	1783	2189	1783	2189	1783	2189
PopGrowth & construction	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Note: Neighboring capacity p.c. is the sum of PV installations (measured in MWp and normalized by the working-age population) across all other counties in the same spatial planning region (columns (1) and (2)), the 5 closest counties (columns (3) and (4)), or the 10 closest counties (columns (5) and (6)). Closeness is measured by the distance between the counties' most populous municipalities. The *year fixed effects* are estimated at the level of the state × county type (rural or urban county). All other variables are defined as in Table 1. Panel A reports the results for the time series split, and Panel B reports the results for the cross-sectional split. Standard errors (in parentheses) are clustered at the level of 94 spatial planning regions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Spillovers from neighboring counties The second channel, hiring workers from outside the local labor market (a county in our setting), is also unable to fully explain our findings. First, we control for population growth in all of our regressions, which should capture migration as long as workers also change their place of residence. Second, migration and commuting cannot explain the results of our time-series split, given that almost all counties are classified as slack and tight at the same time. Third, we show next that there is no evidence for demand spillovers across regions independent of the state of the labor market, implying that counties in this setting are indeed self-contained labor markets.

To test for geographic spillovers, we follow the approach of Acconcia et al. (2014) and include investments in neighboring counties as an additional control variable. We consider three possible definitions of neighboring counties: all other counties within the same spatial planning region, the five closest counties based on the distance between both counties most populous municipalities, and the ten closest counties. For each set of a countys neighbors we calculate the total PV installations in MWp within the set of neighboring counties and normalize the total installations with the working-age population in the county of interest. Given this, we estimate an extended version of the main empirical model (1) that includes aggregate PV installations in neighboring counties as an additional covariate. As in the main empirical analyses, we classify counties as having slack or tight labor markets according to their own unemployment rate as described in Section 3.2.

Table 5 reports the OLS estimates of the demand spillovers conditional on the state of the labor market as defined via the time series split (Panel A) as well as the cross-sectional split (Panel B). In both splits and in all three definitions of a county's set of neighbors, the effect of additional PV installations in geographically proximate regions is at least one order of magnitude smaller than the effect of additional installations within the county. In addition to their small magnitude, all the coefficients are statistically insignificant. The estimated effects of the demand spillovers also do not differ by much between slack and tight labor markets, while the differences of the employment gains due to the within-county investments remain at the same level as in Table 1, the main OLS specification.[37] All in all, the employment effects of PV installations are very local in nature, so that demand spillovers are largely unimportant for the interpretation of our findings.

Wage growth Taken together, this leaves crowding out as the most plausible mechanism. Workers installing PV systems might be drawn from other jobs if the labor market is tight, while they might have been unemployed if the labor market is slack. This is the mechanism identified by Michaillat (2014), who argues that the employment response to additional demand is a general function of the state of the labor market. In his model, diminishing returns to labor lead to a quasi-

[37] Appendix D.2 shows that there is no evidence for demand spillovers in the full sample either, and Table D.3 in Appendix D.3 demonstrates that the IV estimates lead to the same conclusions as the OLS estimates.

Table 6
Wage growth.

	Δ Log(Median wage in construction)				
		Split along			
		Time series		Cross-section	
	Baseline	Slack	Tight	Slack	Tight
Years	03–12	03–07	08–12	03–12	03–12
	(1)	(2)	(3)	(4)	(5)
Avg. yearly capacity p.c.	9.21**	3.02	11.17***	6.88	10.09**
	(3.53)	(18.44)	(3.15)	(10.27)	(4.05)
Population growth	−0.00	−0.10	0.16**	−0.02	0.02
	(0.05)	(0.08)	(0.07)	(0.12)	(0.08)
Avg. yearly construction p.c.	0.11	0.22	−0.49**	0.42	−0.04
	(0.21)	(0.23)	(0.25)	(0.42)	(0.28)
State × county-type fixed effects	Yes	Yes	Yes	Yes	Yes
P-value slack > tight		0.206		0.190	
Observations	368	363	370	159	209

Note: The dependent variable is the difference in the log median wage in the construction sector over the years indicated in the row *"Years"* (referred to as "sample years"). *Avg. yearly capacity p.c.* are the average yearly PV installations during the sample years measured in megawatt peak (MWp) normalized by the working-age population in 2003 (indicated by "p.c." for "per capita"). *Population Growth* is the difference in the working-age population over the sample years relative to the working-age population in 2003. *Avg. yearly construction p.c.* is the average yearly number of residential and non-residential buildings completed during the sample years. For the time-series defintion of slack in columns (2) and (3) we split the sample into the years 2003–2007 (slack) and 2008–2012 (tight). In the cross-section in columns (4) and (5), we split the sample relative to the mean of the average unemployment rate at the state-level in 2002. *P-value slack > tight* reports the *p*-value of the test of the null hypothesis that the wage effect of PV installations is larger in slack than in tight labor markets. The number of observations is smaller than the number of counties (400) because the wage data is only available for county-years in which the number of employees in construction exceeds 1000. Standard errors (in parentheses) are clustered at the level of 94 spatial planning regions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

labor supply curve that is convex in labor market tightness, so that additional labor demand leads to a higher degree of crowding out in a tight than in a slack labor market.

Furthermore, in this model wages respond stronger to a given demand shock in tight than in slack labor markets. To test this prediction, we estimate how wage growth reacts to PV installations using the following model:

$$\frac{[log(wage_{c,t}) - log(wage_{c,t-k})]}{k} = \beta \cdot \frac{1}{k} \cdot \sum_{\tau=0}^{k-1} PV\ Installations\ p.c._{c,t-\tau}$$
$$+ \gamma_1 \cdot \frac{1}{k} \cdot \sum_{\tau=0}^{k-1} Construction\ p.c._{c,t-\tau} + \gamma_2 \cdot \frac{Pop_{c,t} - Pop_{c,t-k}}{k \cdot Pop_{2003}} + \delta_c\, \mathbb{1}[State_c \times CountyType_c] + \varepsilon_c. \quad (3)$$

Here, the dependent variable is the average yearly growth of the median wage in construction in county c over k years.[38] The independent variable of interest is the average amount of yearly PV installations (in per-capita terms) during the same years. We also control for average construction activity, the average yearly change in the working age population, and state × county-type fixed effects. As baseline, we estimate model (3) for wage growth between 2003 and 2012 for the full set of counties. For the cross-sectional split, we consider the same time horizon and classify counties as slack if their unemployment rate in 2002 exceeds the state average unemployment rate in 2002. Along the time series, we split the sample into the periods of 2003–2007 (slack) and 2008–2012 (tight).[39]

This model builds on Autor et al. (2013) and evaluates the effect of PV installations on wage growth in the medium run. We focus on the medium run to account for the common finding of sticky wages (Taylor, 2016). Sticky wages are particularly relevant in the context of PV installations as almost 90% of workers in the construction sector are subject to collective bargaining agreements on the industry and regional level, which on average have a duration of two years.[40] Furthermore, bargaining outcomes may not immediately adjust to demand shocks.[41]

Table 6 shows the results of estimating Eq. (3) via OLS. For the full set of counties, there is a statistically significant response of wage growth to additional PV installations (column (1)). The coefficient of 9.21 implies that a one standard de-

[38] Due to privacy requirements, we can only study local wages at the one-digit industry level. This is a coarser level than the "high exposure" sector in Table 4, which is a collection of three-digit industries. We focus on wage growth in the construction sector, the one-digit industry with the largest share of employment classified as high exposure.

[39] The cross-sectional split corresponds to row (10) of Table 3 and the time series split to row (2). Both splits ensure that the classification of slack and tight markets is constant for the considered horizons.

[40] For durations of collective bargaining agreements, see https://www.boeckler.de/wsi-tarifarchiv_4832.htm, last accessed 11/25/2018.

[41] The survey evidence by Smets and Lamo (2009) shows that the wage elasticity to idiosyncratic shocks is low and that firms are more likely to adjust employment than wages in response to demand shocks. For PV installations, this may be particularly relevant as it was not clear ex ante how permanent these shocks would be.

viation increase in capacity leads to a yearly increase in the median construction wage of 0.05 percentage points. The splits in tight and slack labor markets reveal that this additional wage growth is driven almost exclusively by tight markets, where the coefficients are around 10 as well. In contrast, the coefficients for slack markets are small and imprecisely estimated. For this reason, we cannot reject the null hypothesis that the effect of PV installations on wage growth is larger in slack than in tight markets on conventional levels. Overall, these results mirror the findings of the employment gains and point towards crowding out as an explanation.[42]

6. Conclusion

In conclusion, this paper shows that the employment effects of PV investment in Germany are state-dependent. It overcomes all three challenges to the estimation of state dependent multipliers: First, we study an exogenous demand shock, as PV investment was mainly driven by factors independent of regional labor market dynamics. Second, this demand shock provides ample variation in all states of the economy as more than € 60 billion were invested across 400 counties over ten years. Third, the composition of investments has been constant as each PV installation of a given size constituted the same shock. This allows us to compare the estimated multipliers between the different states of the economy.

Our results are consistent with crowding-out being responsible for the low impact of PV installations on employment in tight labor markets. Thus, our paper adds evidence for a particular mechanism underlying state-dependence. Tracing out the mechanism is a plausible way forward in studying state-dependent multipliers as it might help to reconcile the different results in the literature. This paper takes a first step in this direction.

Finally, our paper has two policy implications: First, investments during a recession pay a double dividend as they put additional people to work, while they mostly lead to crowding out in booms. Hence, economic downturns are a good time to undertake public investment programs. Second, place-based policies provide a better return in terms of jobs in regions with high unemployment than with low unemployment. This suggests that the design of make-work programs should take economic circumstances of targeted areas into account, even if we ignore all equity concerns.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jmoneco.2019.02.006.

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[42] Table D.5 in Appendix D.4 shows the corresponding IV results which have the same qualitative pattern.

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